Chapter5: Electric Field and Gauss' Law

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 9th edition]

Topics to be covered:

22-2: The Electric Field, 22-3: Electric Field lines, 22-4: The Electric Field Due to a Point Charge, 22-5: The Electric Field Due to an Electric Dipole, 22-6: The Electric Field Due to a Line of Charge,22-7: The Electric Field Due to a Charged Disk, 23-2: Flux, 23-3: Flux of an Electric Field, 23-4: Gauss' Law

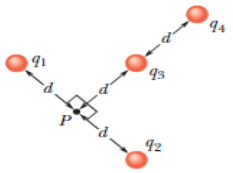
Theory:

1. Define electric field and electric field line.
2. Derive the expression for the magnitude and direction of electric field due to a point charge.
3. What are electric dipole and electric dipole moment?
4. Find out the expression for the electric field due to an electric dipole along its axis.
5. Consider a positively charged ring of radius *R*. Find the electric field at a point, a distance *z* from the plane of the ring along its central axis.
6. Define electric flux and explain Gauss' Law.

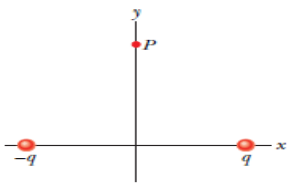
Problems:

Chapter-22

1. [5] What is the magnitude of a point charge whose electric field 50 cm away has the magnitude 2.0 N/C?
2. [4] Two particles are attached to an x axis: particle 1 of charge -2.00×10^{-7} C at x = 6.00 cm, particle 2 of charge $+2.00\times10^{-7}$ C at x =21.0 cm. Midway between the particles, what is their net electric field?
3. [8] In the figure, the four particles are fixed in place and have charges $q_1=q_2= +5e$, $q_3=+3e$, *and* $q_4= -12e$. Distance $d= 5.0\ \mu m$. What is the magnitude of the net electric field at point P due to the particles?



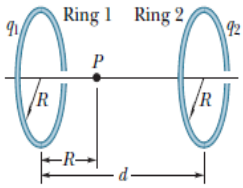
4. [9] Figure shows two charged particles on an x axis: q =-3.20×10^{-19} C at x =-3.00 m and q = 3.20×10^{-19} C at x =+3.00 m. What are the (a) magnitude of the net electric field produced at point P at y = 4.00 m?



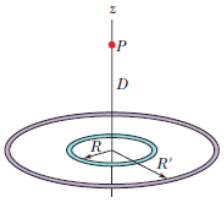
5. [80] Calculate the electric dipole moment of an electron and a proton 4.30 nm apart.

6. [Sample Problem] Assume a vertical electric dipole that has charge -*q* at cloud height *h* and charge +*q* at below ground depth *h*. If *q*= 200 C and *h*= 6.0 km, what is the magnitude of the dipole's electric field at altitude *z*=30 km somewhat above the clouds?

7. [23] Figure shows two parallel non-conducting rings with their central axes along a common line. Ring 1 has uniform charge q_1 and radius *R*; ring 2 has uniform charge q_2 and the same radius *R*. The rings are separated by distance *d =3.00R*. The net electric field at point *P* on the common line, at distance *R* from ring 1, is zero. What is the ratio q_1/q_2?

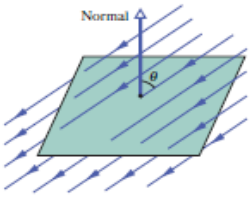


8. [30] The figure shows two concentric rings, of radii *R* and *R' = 3.00R*, that lie on the same plane. Point *P* lies on the central *z* axis, at distance *D=2.00R* from the center of the rings. The smaller ring has uniformly distributed charge *+Q*. In terms of *Q*, what is the uniformly distributed charge on the larger ring if the net electric field at *P* is zero?



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9. [1] The square surface shown in Figure measures 3.2mm on each side. It is immersed in a uniform electric field with magnitude *E = 1800 N/C* and with field lines at an angle of *Θ= 35°* with a normal to the surface, as shown. Take that normal to be directed "outward," as though the surface were one face of a box. Calculate the electric flux through the surface.



10. [7] A point charge of 1.8 µC is at the center of a Gaussian cube 55 cm on edge. What is the net electric flux through the surface?